



**FINDING AND LOCATING A PERSON OR THING
BY USING THE BEACON SYSTEM**

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BORRIHAN BOONRUANG

**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2020

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BORRIHAN BOONRUANG

**A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
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IN COMPUTER ENGINEERING**

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2020

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ABSTRACT

Beacon is an advanced IoT or Internet of Things technology that has been popular abroad for a while. The work of the Beacon is not as complicated as you think. The principle of operation is similar to that of RFID technology, but the beacon uses a Bluetooth transmission method. Which sends a low energy signal, known as Bluetooth 4.0 Low Energy at 2.4 GHz frequency to a device. Recipient automatically to the recipient's device. It will have its working radius here. if it falls out of radius. it will not be able to work itself. The beacon itself can be very useful for use in areas that GPS can't reach. Such as in the mall or in buildings.

The introduction of Beacon Technology into serious use in Thailand is beginning to be seen more and more. Especially used in areas where the GPS system is not accessible. Such as indoors. The Beacon Technology device will send the Bluetooth signal to the user in the radius that the signal is transmitted. After that, it will Notifications up and transfer data.

Technically, Beacon Technology can also be applied in many other applications. Including office work, shop front, or even logistics. It can also use Beacon Technology to develop models to be able to use this technology. However, the most important thing is event planning. Especially a team of innovators, application developers and marketers. These two teams must think and work together very well. This is in order to reduce errors and achieve effective innovation

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CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Nowadays, in our life have many activities to do. From time to time, we focus on those activities and forget to notice that our stuff is still with us or not. When we realized that the things had disappeared. we may forget those items somewhere near us, but we can't find them and can't imagine where they are.

According to the Wall Street Journal show a survey by British insurers published 2 years ago. Indicates that the average person is forgotten about 9 times a day, and 1 in 3 respondents say that each day they take about 15 minutes on average to find your missing item. Mobile, phones, keys and documents are at the top of the list of things humans tend to forget the most. Sometimes those problems can be frustrating because we want to use them, but we can't find them, and this problem applies to people of all genders and ages.

So that we came up with an idea for solving these problems using a technology called beacon, using it as a signaling link to determine location.

1.2 Objective

To develop a system called beacons for solving problems we need to find necessary items that we don't remember where we placed or stored.

1.3 Scope

1.3.1 The system can notify you when the device is connected.

1.3.2 The system is connected to determine the location of the devices in the area where the beacon is installed.

1.3.3 The system can record the connection history of the device.

1.3.4 At least 3 beacon devices will be used.

1.4 Methodology

The function of this system is to use Bluetooth broadcasting so that devices can connect to each other at specified distances or territories.

1.4.1 Install the beacon in the desired location to receive or transmit signal.

1.4.2 Install the beacon on the object that we want to track.

1.4.3 The devices will be connected when entering the broadcast range.

1.4.4 Once connected, it will show which beacon the object we tracked is connected.

1.5 Plan

Table 1.1 Project planning

Task / Month	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Finding the topic and the advisor.									
Research the information.									
Start to make project on software and hardware.									
Computer Engineer Pre-project document preparation									
Computer Engineer Pre-project final documentation and presentation									
System Development									
System testing									
Computer Engineer project document preparation									
Computer Engineer project final documentation and presentation									

Table 1.1 Plan.

1.6 Expected Result

The expected result of this project is as follow.

1.5.1 The beacon system is developed.

1.5.2 This beacon system can used to find each other in the broadcast range.

1.7 Resources

1.7.1 Equipment

1.7.1.1 Hardware

- ESP 32
- ESP 32 with OLED display
- Laptop

1.7.1.2 Software

- Arduino IDE

1.7.2 Place

- Mae Fah Luang University

CHAPTER 2

LITERATURE REVIEW

2.1 Beacon

Beacon is a technology in the IOT or Internet of Things era that has been popular abroad since 2013, especially in the retail industry and the museum. The Beacon can present information to you as you walk closer to an object, for example, you will see a promotion. New features when you open the store door. Or know details about a painting you are viewing in an art gallery, etc. This astonishing ability has enabled many hardware startups, both in Thailand and abroad, to develop their own Beacons into various products.

In this project, we use Beacon, RSSI technology, to know the value used to tell the strength of the received radio signal in terms of power in dBm, where the RSSI value varies with the signal strength.



Figure 2.1 BLE Beacon.

2.2 Related Work

2.2.1 Bluetooth

Using Bluetooth as a connection to the operation of the system. Bluetooth uses a high frequency 2.4 GHz radio signal (gigahertz), but is broken down. According to each country for example, in Europe and America it uses the 2.400 to 2.4835 GHz range. It is divided into 79 channels and will use this divided channel to send data, switching channels back and forth 1,600 times a second, the Japanese used frequency 2.402 to 2.480 GHz. Divided into 23 channels, the Bluetooth operating range is 5-100 meters with a protection system using input Code before connection and prevent interception during communication Which the system will switch the channel back and forth Will have the ability to automatically change the frequency of the contact. Without having to sort by channel number Makes wiretapping or stealing more difficult, primarily of Bluetooth is designed for use with smaller devices. Due to the transport of small amounts of data such as files, images, audio, applications. And can be easily moved It is only within the specified distance (about 5-100 meters). It is also low power consumption, and can be used for a long time. Without having to bring it to charge frequently too.



Figure 2.2 How Bluetooth work.

2.3 Technology and Tools

2.3.1 Bluetooth low energy

Bluetooth low energy (BLE) is a feature of Bluetooth 4.0 technology aimed at the next generation of low energy, low latency wireless devices within short distances (up to 50 - 160 m, 50 m / 160) This specification facilitates a wide range of applications and small equipment used in healthcare, fitness, security applications. And the home entertainment industry

A device that uses Bluetooth low energy wireless technology. It is expected to consume very little power compared to traditional Bluetooth devices. Will enable a number of products to communicate via Bluetooth. In many cases, products can be operated for more than a year using just a button cell without recharging. It is therefore possible to build a measuring device, such as a thermometer, that works continuously and at the same time communicating with other devices. Such as a mobile phone This may raise concerns over privacy concerns, as having a low-power, continuous telemetry device takes advantage of it or similar.

Please note that the low power consumption of the device is not affected by its behavior when transmitting radio waves. This is due to the protocol design so that it can have a low duty cycle while considering the use cases of Bluetooth Low Energy (BLE) devices when applied. For continuous data transfer, it has a lower power consumption rate than normal Bluetooth devices that transmit the same amount of data. And in fact, the device tends to use more power Because this protocol is suitable for the transmission of clusters of information for a short period of time.

2.3.2 RSSI (Receive Signal Strength Indicator)

RSSI, short for Receive Signal Strength Indicator, is a value used to determine the strength of a received radio signal in terms of power. It is measured in dBm (decibels milliwatt), where the RSSI value varies with the signal strength. That is, if the RSSI value is large, the signal is received with a high strength. That is, the transmitter and receiver are close to each other and vice versa, if the RSSI value is small, the signal received is low, the transmitter and receiver are far apart.

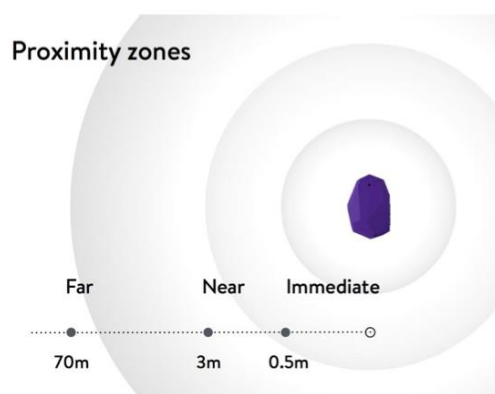


Figure 2.3 Receive Signal Strength Indicator.

2.3.3 ESP32

ESP32 is the name of an IC microcontroller that supports built-in Wi-Fi and Bluetooth 4.2 BLE connectivity, manufactured by Espressif.

There is a detailed specification as follows.

- The CPU uses a 2-core Tensilica LX6 architecture, the brain clock is 240MHz
- Has 512KB built-in RAM.
- Support for connecting external ROM up to 16MB.
 - It comes with 802.11 b / g / n Wi-Fi and supports both Station softAP and Wi-Fi direct modes.
 - Built-in Bluetooth It supports operation in 2.0 mode and 4.0 BLE mode.
 - Operating voltage 2.6V to 3V is used.
 - Works at temperatures from -40 C to 125 C.

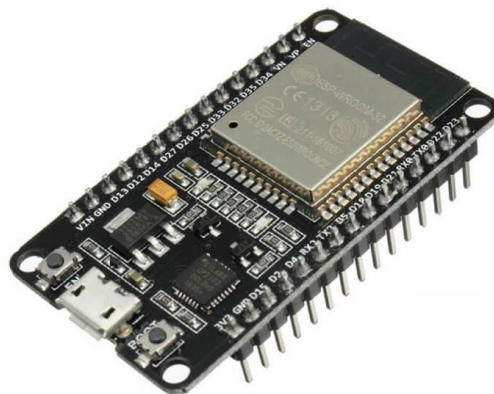


Figure 2.4 ESP32

2.3.4 ESP32 + OLED 128x64 on one board

ESP32 Development Board with OLED Display With the growing popularity of the ESP32, the development of the development board is more diverse. For this development board It has the advantage that it combines an ESP32 and an OLED display on a single board. Making it easier to use. The board supports direct program uploading via computer using a USB to UART chip, CP2102 number for Reset switch and mode selection are below the plate.

There is a detailed specification as follows.

- Use IC chip USB to UART No. CP2102.
- There is a circuit into automatic program upload mode, arranged in a node MCU circuit.
- Use IC regulator AMS1117-3.3 to convert 5V voltage from USB to 3.3V to supply ESP32 total current up to 1A.
- There are switches to select the active mode (Boot) and Reset (EN) at the bottom.
- Connect to OLED display via I2C at position 0x3C and connect to GPIO5 (SDA) and GPIO4 (SCL) pins.

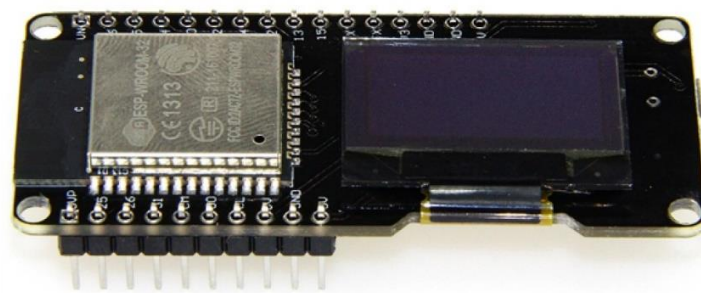


Figure 2.5 ESP32 + OLED 128x64 on one board

2.3.5 IFTTT

If This Then That (commonly known as IFTTT) is a service that allows a user to program a response to events in the world of various kinds. There is a long list of kinds of events to which IFTTT can respond, all detectable via the Internet. An example event is that Weather Underground reports rain is forecast for tomorrow. Another is that someone tagged the user in a photo on Facebook. There is similarly a long list of kinds of responses that are possible, all executable via the Internet. An example response would be to send an email to the user saying rain is forecast or copy the aforementioned photo to the user's archive.

IFTTT has partnerships with hundreds of service providers that supply event notifications to IFTTT and execute commands that implement the responses, but some event and command interfaces are just public APIs.

The programs, called applets, are simple and created graphically.



Figure 2.6 IFTTT

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Activity Diagram

The analysis of the system is demonstrated by an activity diagram.

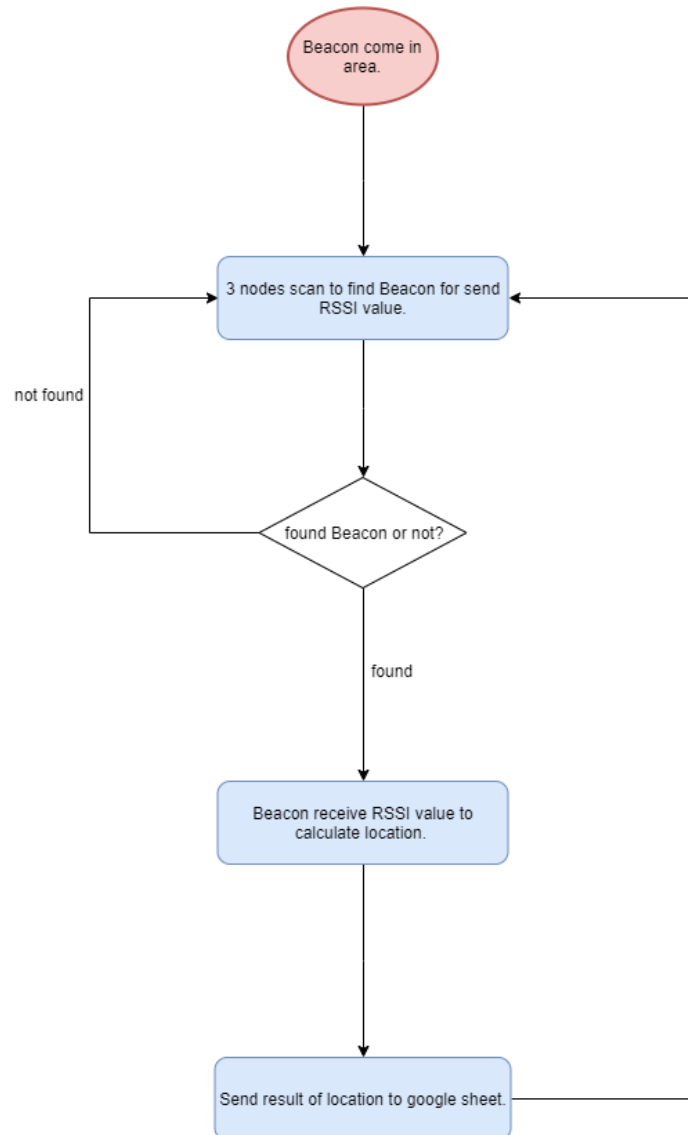


Figure 3.1 Activity Diagram.

The operation starts by 3 nodes scan to find Beacon device to send RSSI value. Beacon device will keep RSSI value to calculate to find location and send location to save history in Google sheet.

3.2 Use Case Diagram

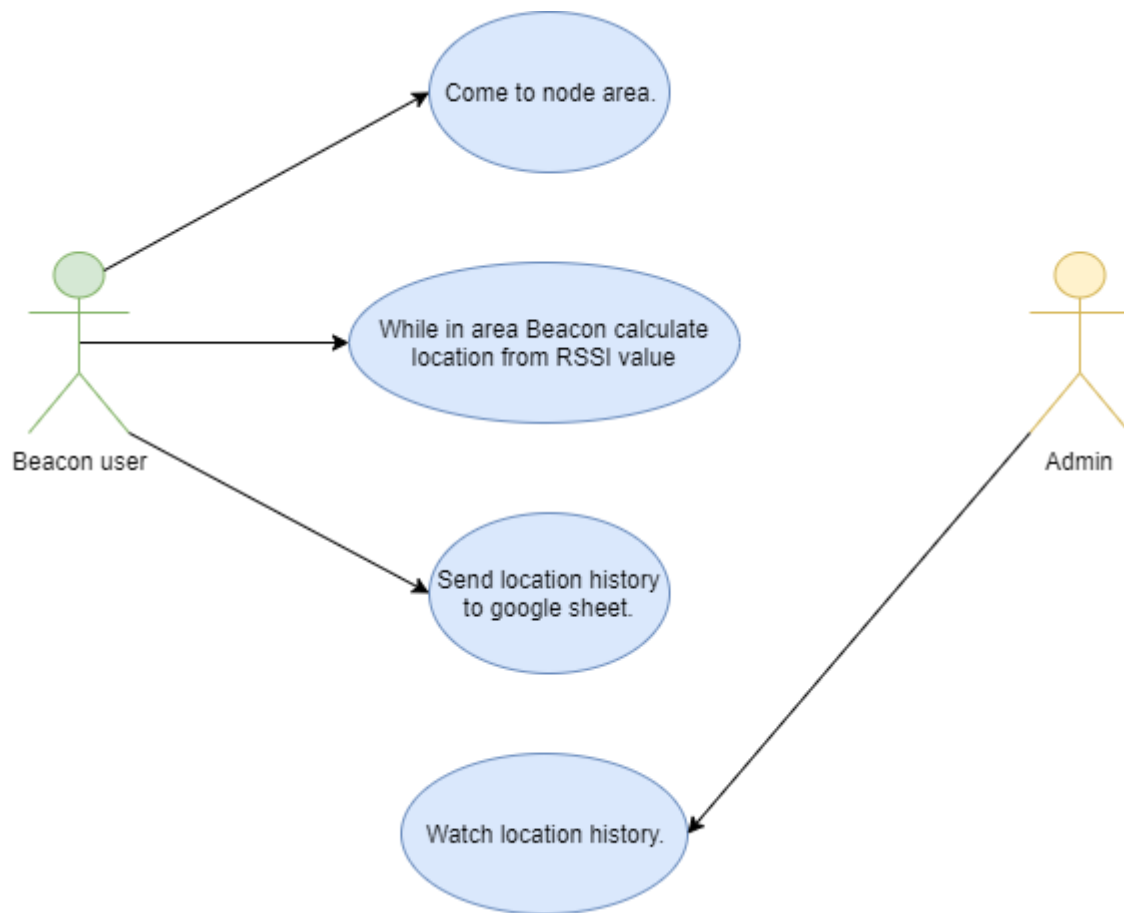


Figure 3.2 Use case diagram.

When the Beacon device user enters or exits the node's transmitting range. It can send a RSSI value to Beacon device. After, calculate RSSI value from 3 nodes Beacon device can send location to Google sheet for keep in history. Admin can see where is Beacon device in that moment or whatever time of Beacon device while stay in area.

CHAPTER 4

EXPERIMENT AND RESULTS

From the experimental perspective, we use 3 esp32 set to be BLE node and 1 esp32 set for Beacon device. we place 3 esp32 in triangle to set area for scan to find and send RSSI value to Beacon device. we set 1 esp32 for Beacon device to receive RSSI value for calculate to find position in area.

Moreover, we use API from IFTTT called webhook. We use this API service because it can easily connect between Beacon device and Google sheet together. We do like this for keep the calculate position from Beacon device in history and can check it after.

4.1 Beacon system

We developed this system for make a device to tracking. Beacon device must determine MAC address to let node know for done the all process. After, all process done we can check position of Beacon device in scan history in Google sheet.

4.1.1 Sender

We use 3 nodes to be a RSSI sender for set triangle boundary area. We determine the position in area by set RSSI value about -30, -60, -90 dM. When Beacon come in the process will start.

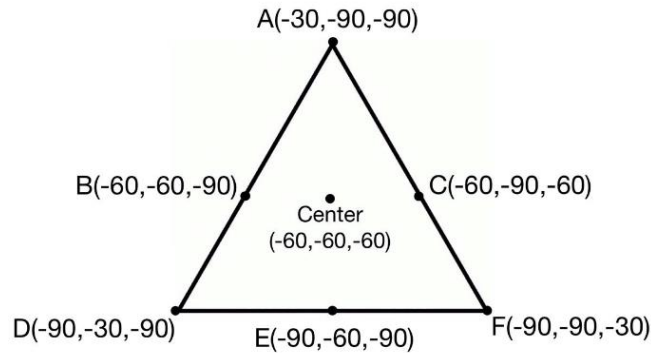


Figure 4.1 triangle boundary area.

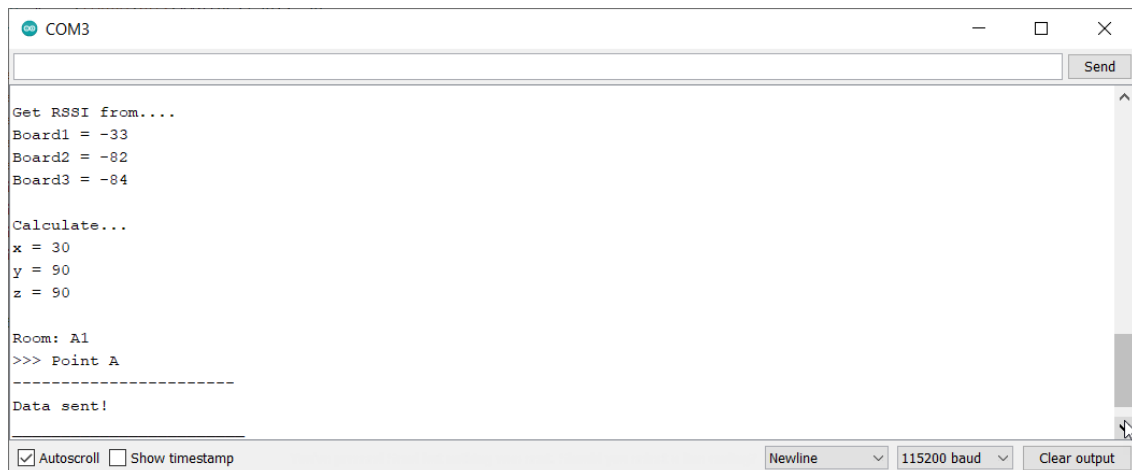
4.1.2 Receiver

When Beacon device come in area. Node will send RSSI value to Beacon device. If Beacon device receive RSSI value not more than -45 dM. Beacon Device will calulate to set RSSI value to -30 dM. If Beacon device receive RSSI value around -45 and -74 dM. Beacon Device will calulate to set RSSI value to -60 dM. If Beacon device receive RSSI value more than -75 dM. Beacon Device will calulate to set RSSI value to -90 dM. When Beacon device receive RSSI value from 3 nodes. It can calculate to find position in triangle area and send data to Google sheet.

4.2 Results

From the results, When Beacon device receive RSSI value from all 3 nodes. It will perform process to calculate and send position data. When done all process will get results like this.

4.2.1 When Beacon device get RSSI value around -30, -90, -90. The result will show point A.

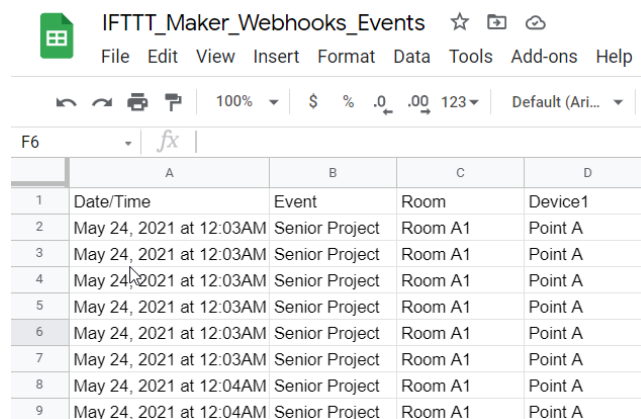


```
COM3
Get RSSI from....
Board1 = -33
Board2 = -82
Board3 = -84

Calculate...
x = 30
y = 90
z = 90

Room: A1
>>> Point A
-----
Data sent!
```

Figure 4.2 Calculation from Beacon device to show point A.



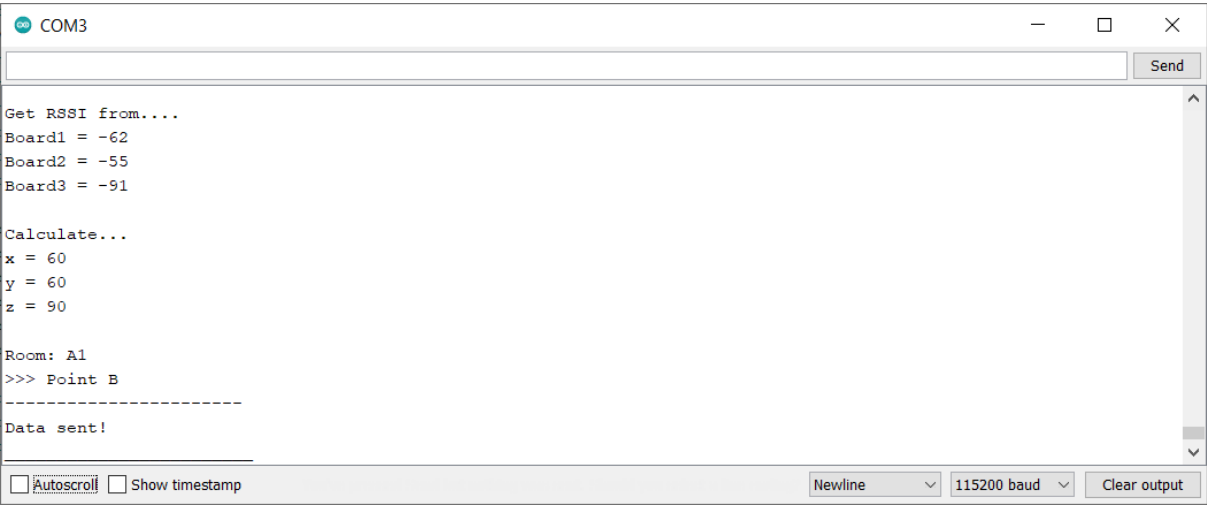
	A	B	C	D
1	Date/Time	Event	Room	Device1
2	May 24, 2021 at 12:03AM	Senior Project	Room A1	Point A
3	May 24, 2021 at 12:03AM	Senior Project	Room A1	Point A
4	May 24, 2021 at 12:03AM	Senior Project	Room A1	Point A
5	May 24, 2021 at 12:03AM	Senior Project	Room A1	Point A
6	May 24, 2021 at 12:03AM	Senior Project	Room A1	Point A
7	May 24, 2021 at 12:03AM	Senior Project	Room A1	Point A
8	May 24, 2021 at 12:04AM	Senior Project	Room A1	Point A
9	May 24, 2021 at 12:04AM	Senior Project	Room A1	Point A

Figure 4.3 Send point A data to Google sheet.



Figure 4.4 display show point A on Beacon device.

4.2.2 When Beacon device get RSSI value around -60, -60, -90. The result will show point B.

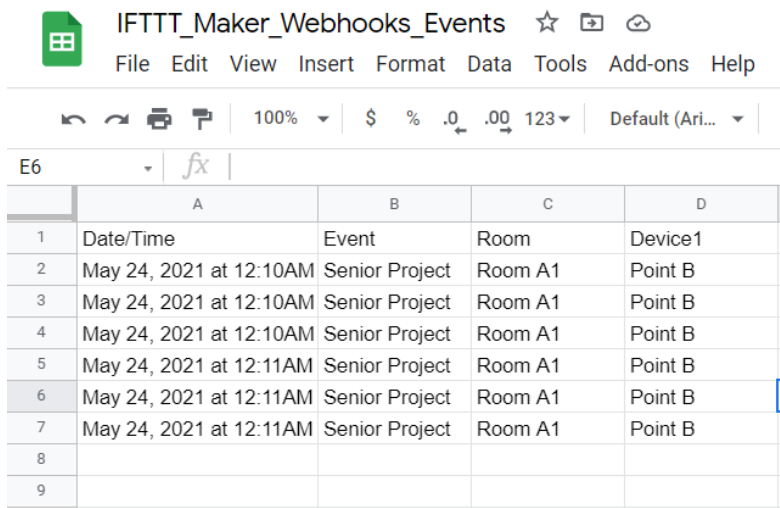


```
COM3
Get RSSI from....
Board1 = -62
Board2 = -55
Board3 = -91

Calculate...
x = 60
y = 60
z = 90

Room: A1
>>> Point B
Data sent!
```

Figure 4.5 Calculation from Beacon device to show point B.



	A	B	C	D
1	Date/Time	Event	Room	Device1
2	May 24, 2021 at 12:10AM	Senior Project	Room A1	Point B
3	May 24, 2021 at 12:10AM	Senior Project	Room A1	Point B
4	May 24, 2021 at 12:10AM	Senior Project	Room A1	Point B
5	May 24, 2021 at 12:11AM	Senior Project	Room A1	Point B
6	May 24, 2021 at 12:11AM	Senior Project	Room A1	Point B
7	May 24, 2021 at 12:11AM	Senior Project	Room A1	Point B
8				
9				

Figure 4.6 Send point B data to Google sheet.



Figure 4.7 display show point B on Beacon device.

4.2.3 When Beacon device get RSSI value around -60, -90, -60. The result will show point C.

```

COM3

Get RSSI from....
Board1 = -70
Board2 = -91
Board3 = -62

Calculate...
x = 60
y = 90
z = 60

Room: A1
>>> Point C
-----
Data sent!

Autoscroll Show timestamp Newline 115200 baud Clear output

```

Figure 4.8 Calculation from Beacon device to show point C.

	A	B	C	D
1	Date/Time	Event	Room	Device1
2	May 24, 2021 at 12:19AM	Senior Project	Room A1	Point C
3	May 24, 2021 at 12:20AM	Senior Project	Room A1	Point C
4	May 24, 2021 at 12:20AM	Senior Project	Room A1	Point C
5	May 24, 2021 at 12:20AM	Senior Project	Room A1	Point C
6	May 24, 2021 at 12:20AM	Senior Project	Room A1	Point C
7	May 24, 2021 at 12:20AM	Senior Project	Room A1	Point C
8				
9				

Figure 4.9 Send point C data to Google sheet.

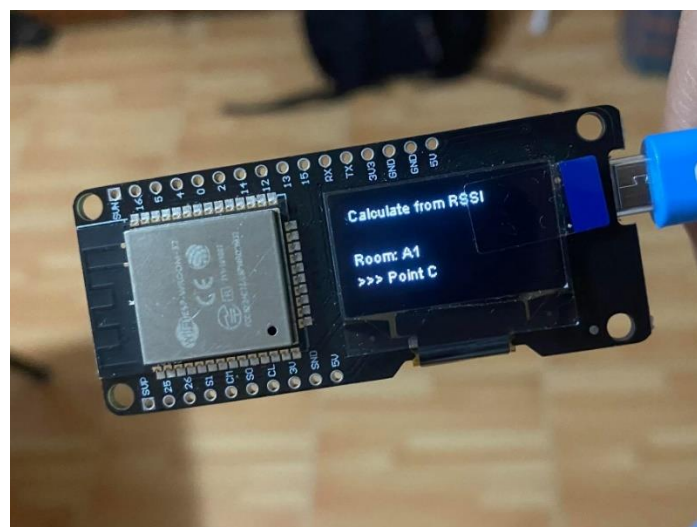
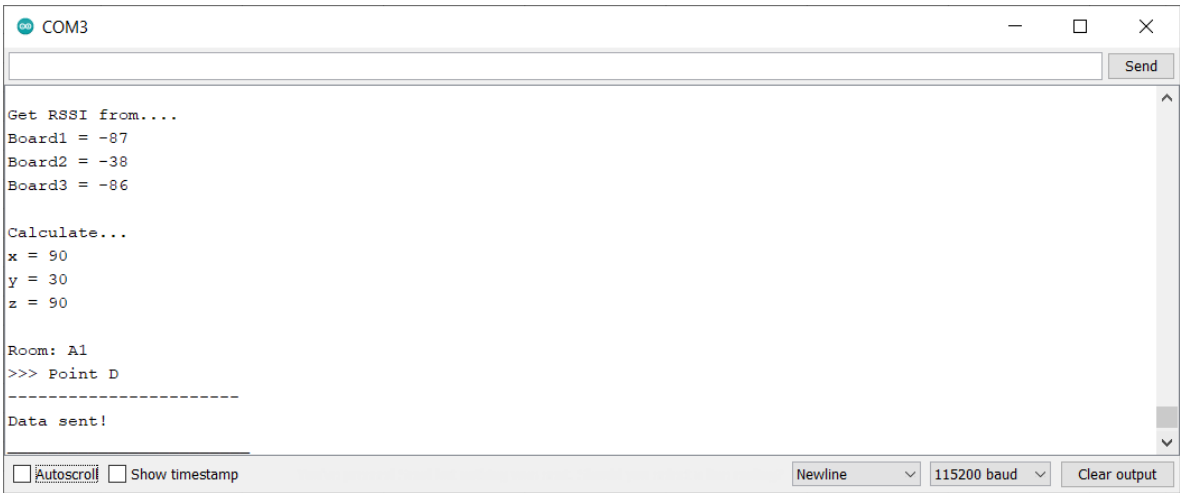


Figure 4.10 display show point C on Beacon device.

4.2.4 When Beacon device get RSSI value around -90, -30, -90. The result will show point D.



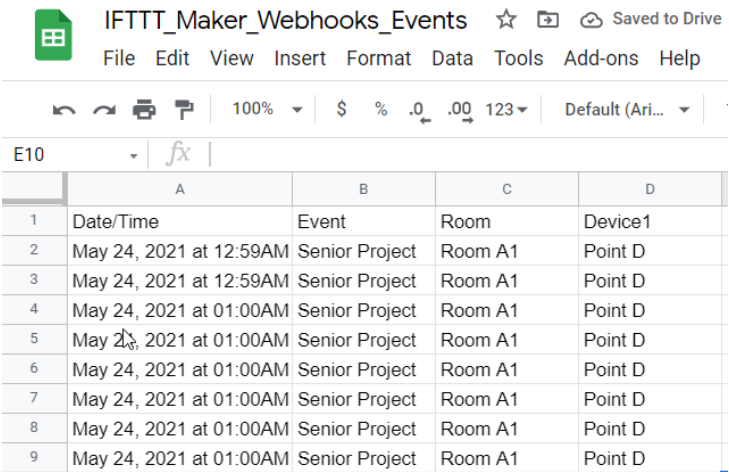
```
COM3
Get RSSI from....
Board1 = -87
Board2 = -38
Board3 = -86

Calculate...
x = 90
y = 30
z = 90

Room: A1
>>> Point D

Data sent!
```

Figure 4.11 Calculation from Beacon device to show point D.



	A	B	C	D
1	Date/Time	Event	Room	Device1
2	May 24, 2021 at 12:59AM	Senior Project	Room A1	Point D
3	May 24, 2021 at 12:59AM	Senior Project	Room A1	Point D
4	May 24, 2021 at 01:00AM	Senior Project	Room A1	Point D
5	May 24, 2021 at 01:00AM	Senior Project	Room A1	Point D
6	May 24, 2021 at 01:00AM	Senior Project	Room A1	Point D
7	May 24, 2021 at 01:00AM	Senior Project	Room A1	Point D
8	May 24, 2021 at 01:00AM	Senior Project	Room A1	Point D
9	May 24, 2021 at 01:00AM	Senior Project	Room A1	Point D

Figure 4.12 Send point D data to Google sheet.

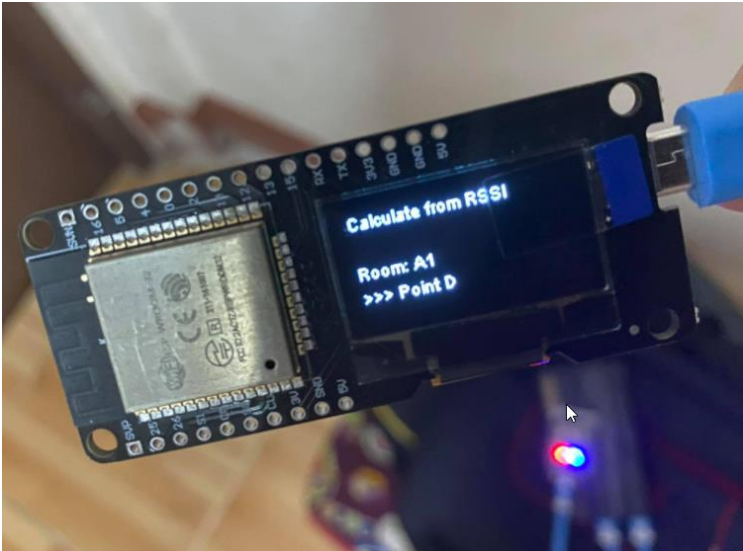


Figure 4.13 display show point D on Beacon device.

4.2.5 When Beacon device get RSSI value around -90, -60, -60. The result will show point E.

```

COM3

Get RSSI from....
Board1 = -89
Board2 = -67
Board3 = -71

Calculate...
x = 90
y = 60
z = 60

Room: A1
>>> Point E
Data sent!

Autoscroll Show timestamp Newline 115200 baud Clear output

```

Figure 4.14 Calculation from Beacon device to show point E.

	A	B	C	D
1	Date/Time	Event	Room	Device1
2	May 24, 2021 at 12:37AM	Senior Project	Room A1	Point E
3	May 24, 2021 at 12:37AM	Senior Project	Room A1	Point E
4	May 24, 2021 at 12:37AM	Senior Project	Room A1	Point E
5	May 24, 2021 at 12:37AM	Senior Project	Room A1	Point E
6	May 24, 2021 at 12:37AM	Senior Project	Room A1	Point E
7				
8				
9				

Figure 4.15 Send point E data to Google sheet.

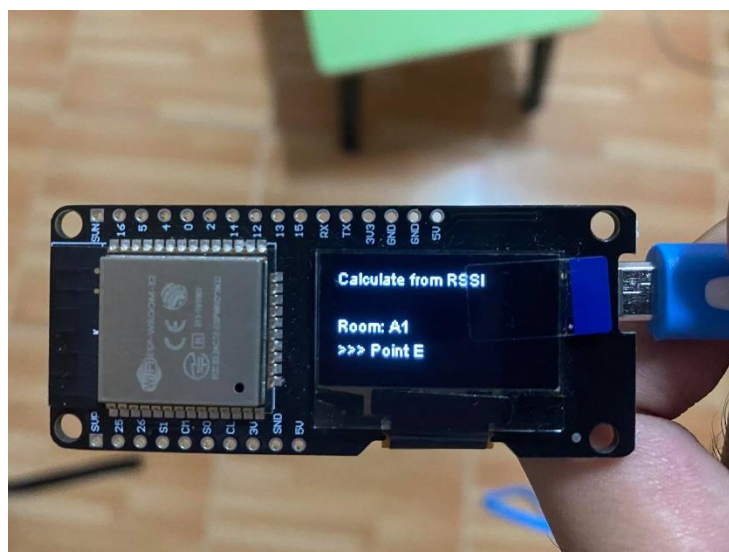
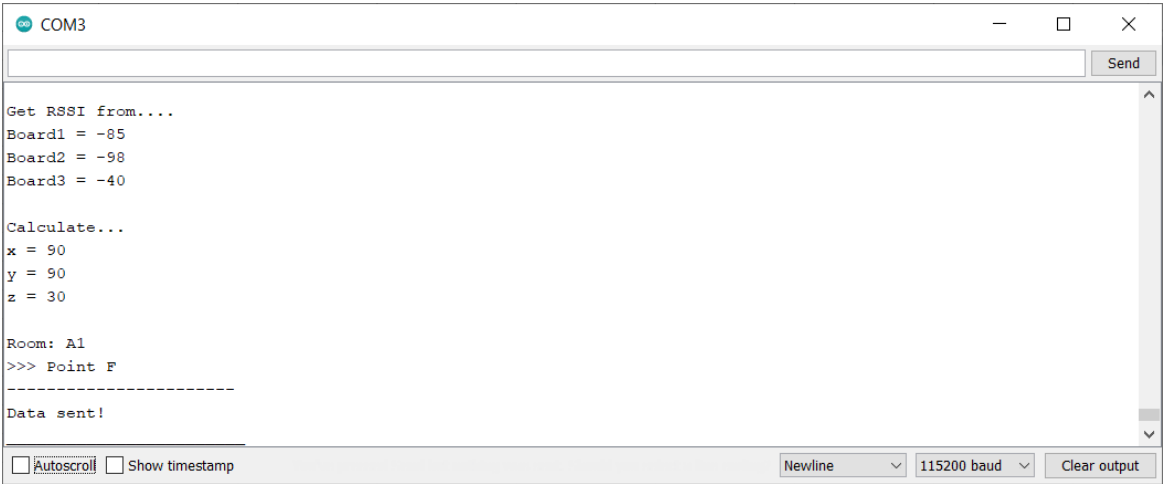


Figure 4.16 display show point D on Beacon device.

4.2.6 When Beacon device get RSSI value around -90, -90, -30. The result will show point F.

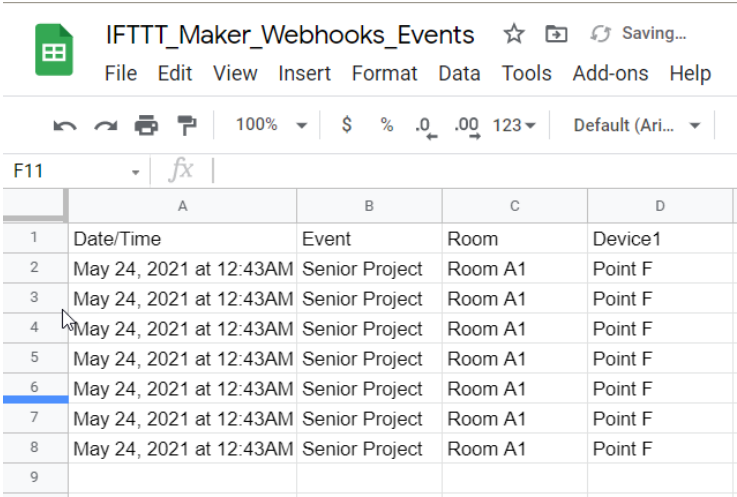


```
COM3
Get RSSI from....
Board1 = -85
Board2 = -98
Board3 = -40

Calculate...
x = 90
y = 90
z = 30

Room: A1
>>> Point F
Data sent!
```

Figure 4.17 Calculation from Beacon device to show point F.



	A	B	C	D
1	Date/Time	Event	Room	Device1
2	May 24, 2021 at 12:43AM	Senior Project	Room A1	Point F
3	May 24, 2021 at 12:43AM	Senior Project	Room A1	Point F
4	May 24, 2021 at 12:43AM	Senior Project	Room A1	Point F
5	May 24, 2021 at 12:43AM	Senior Project	Room A1	Point F
6	May 24, 2021 at 12:43AM	Senior Project	Room A1	Point F
7	May 24, 2021 at 12:43AM	Senior Project	Room A1	Point F
8	May 24, 2021 at 12:43AM	Senior Project	Room A1	Point F
9				

Figure 4.18 Send point F data to Google sheet.



Figure 4.19 display show point F on Beacon device.

4.2.7 When Beacon device get RSSI value around -90, -90, -30. The result will show center point.

```

COM3
Get RSSI from....
Board1 = -56
Board2 = -61
Board3 = -72

Calculate...
x = 60
y = 60
z = 60

Room: A1
>>> Center Point
-----
Data sent!
  
```

Autoscroll Show timestamp Newline 115200 baud Clear output

Figure 4.20 Calculation from Beacon device to show center point.

IFTTT_Maker_Webhooks_Events

	A	B	C	D
1	Date/Time	Event	Room	Device1
2	May 23, 2021 at 11:58PM	Senior Project	Room A1	Center Point
3	May 23, 2021 at 11:58PM	Senior Project	Room A1	Center Point
4	May 23, 2021 at 11:58PM	Senior Project	Room A1	Center Point
5	May 23, 2021 at 11:59PM	Senior Project	Room A1	Center Point
6	May 23, 2021 at 11:59PM	Senior Project	Room A1	Center Point
7	May 23, 2021 at 11:59PM	Senior Project	Room A1	Center Point
8				
9				

Figure 4.21 Send center point data to Google sheet.



Figure 4.22 display show center point on Beacon device.

2.4.8 When don't have Beacon device in area.

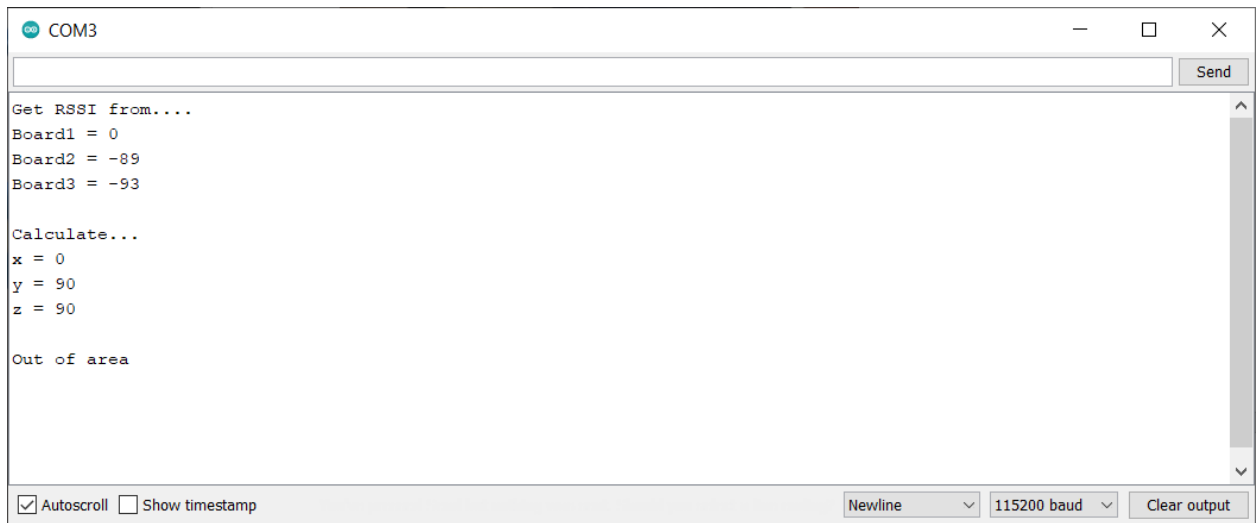


Figure 4.23 Calculation when don't have Beacon device in area.

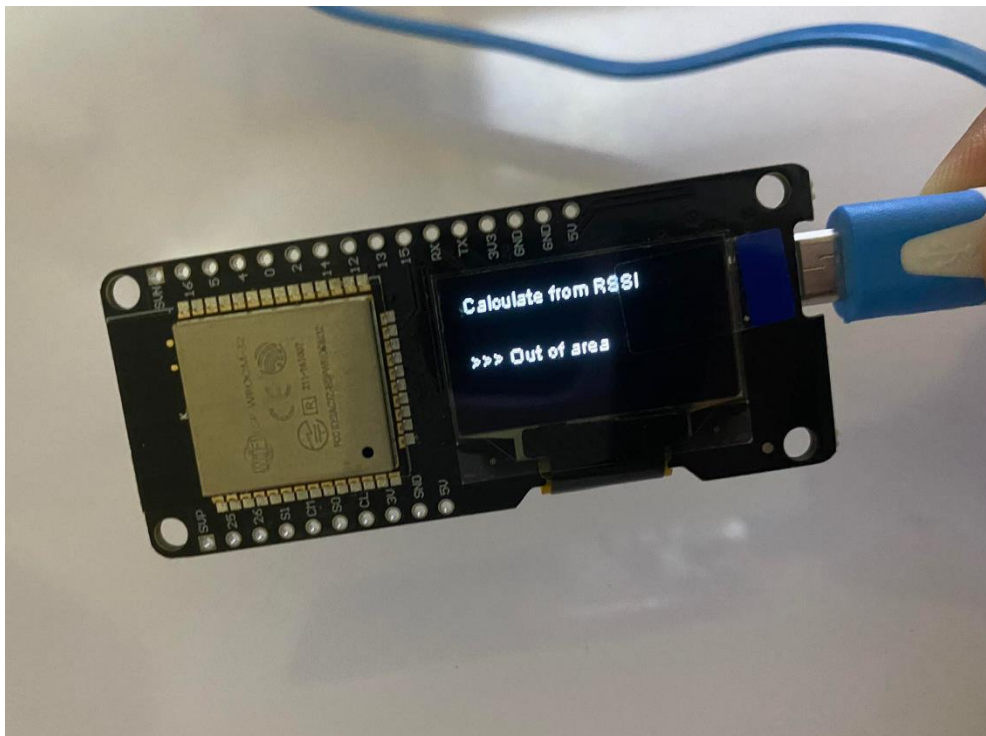


Figure 4.24 display when don't have Beacon device in area.

CHAPTER 5

CONCLUSION AND DISCUSSION

5.1 Conclusion

Beacons are technology in the IOT or Internet of Things era that has been popular abroad since 2013. Beacons is a user location detect technology using location distance measurement (Proximity). Beacons can tell you whether you are near or far from the device. Therefore, Beacons are used in indoor or micro-location based because inside the building, Beacons will determine the user's location more accurately than GPS. There is also a power issue. Beacons use Bluetooth 4.0 LE (Low Energy) to communicate with mobile devices. Bluetooth 4.0 LE is preferred for IoT devices because it consumes less power, unlike GPS that consumes more power. So, this project uses Beacons technology to locate devices in confined spaces in different rooms of the building. It is also possible to save the location history. Make you can know the location of the device at any given time.

5.2 Discussion

Experimenting with this beacon system, due to the limitation of the ESP32's performance. The area determine could not be very broad. Occasionally, scanning for devices to send RSSI values becomes unstable. It makes error to locate the position can occur. When there is an instability in the RSSI transmission. It takes several times to wait for a new transmission to be able to determine the correct location. Because of that, it's cannot send the correct RSSI value in Every time. In addition, a further development of this limitation needs to be corrected in the future.

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